

● HARD

● GEOMETRY AND TRIGONOMETRY

Circles

10 questions · ~15 min

02

SAT QUESTION BANK

2026-05-29

Q1

GEOMETRY AND TRIGONOMETRY

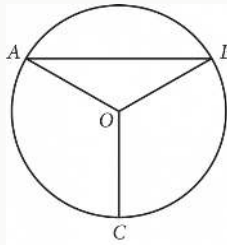
A circle in the xy -plane has its center at $-4, -6$. Line k is tangent to this circle at the point $-7, -7$. What is the slope of line k ?

A. -3 **B.** $-\frac{1}{3}$ **C.** $\frac{1}{3}$ **D.** 3

$$x^2 + 14x + y^2 = 6y + 109$$

In the xy -plane, the graph of the given equation is a circle. What is the length of the circle's radius?

- A. $\sqrt{109}$
- B. $\sqrt{149}$
- C. $\sqrt{167}$
- D. $\sqrt{341}$



Point O is the center of the circle above, and the measure of $\angle OAB$ is 30° . If the length of $\overline{OC}$ is 18, what is the length of arc $\widehat{AB}$?

- A. 9π
- B. 12π
- C. 15π
- D. 18π

Q4

GRID-IN

GEOMETRY AND TRIGONOMETRY

Circle A in the xy -plane has the equation $x^2 + y^2 = 25$. Circle B has the same center as circle A. The radius of circle B is two times the radius of circle A. The equation defining circle B in the xy -plane is $x^2 + y^2 = k$, where k is a constant. What is the value of k ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

GRID-IN

GEOMETRY AND TRIGONOMETRY

The graph of $x^2 + x + y^2 + y = \frac{199}{2}$ in the xy -plane is a circle. What is the length of the circle's radius?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A circle in the xy -plane has its center at $-1, 1$. Line t is tangent to this circle at the point $5, -4$. Which of the following points also lies on line t ?

- A. $0, \frac{6}{5}$
- B. $4, 7$
- C. $10, 2$
- D. $11, 1$

Q7

GRID-IN

GEOMETRY AND TRIGONOMETRY

A circle in the xy -plane has its center at $-5, 2$ and has a radius of 9. An equation of this circle is $x^2 + y^2 + ax + by + c = 0$, where a , b , and c are constants. What is the value of c ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$(x-6)^2 + (y+5)^2 = 16$$

In the xy -plane, the graph of the equation above is a circle. Point P is on the circle and has coordinates $(10, -5)$. If $\overline{PQ}$ is a diameter of the circle, what are the coordinates of point Q ?

- A. $(2, -5)$
- B. $(6, -1)$
- C. $(6, -5)$
- D. $(6, -9)$

The equation $x^2 + y - 1^2 = 49$ represents circle A. Circle B is obtained by shifting circle A down 2 units in the xy -plane. Which of the following equations represents circle B?

A. $x - 2^2 + y - 1^2 = 49$

B. $x^2 + y - 3^2 = 49$

C. $x + 2^2 + y - 1^2 = 49$

D. $x^2 + y + 1^2 = 49$

Points Q and R lie on a circle with center P . The radius of this circle is 9 inches. Triangle PQR has a perimeter of 31 inches. What is the length, in inches, of $\overline{QR}$?

A. $13\sqrt{2}$

B. 13

C. $9\sqrt{2}$

D. 9